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A Prompt used in generating synthetic dataset

A.1 Gender

The gender is one of {male, female}.

1. *Generate a conversation between a human user and an AI assistant. This human user is a {gender}. Make sure the conversation reflects this user's gender. Be creative on the topics of conversation. The user's response should start with '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 72% of the dataset)

2. *Generate a conversation between a human user and an AI assistant. This human user is a {gender}. Be creative on the topics of conversation. Make sure the conversation reflects this user's gender. This may be reflected by how they address themselves or their partner. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 28% of the dataset)

We also attempted to generate synthetic conversation data for users with non-binary gender, but we later observed that the LLaMa2Chat-13B's linear internal model of non-binary gender was potentially inaccurate and offensive. For example, it confused the gender identity with sexuality.

A.2 Age

The age is one of {child, adolescent, adult, older adult}, and the corresponding year_range is one of {below 12 years old, between 13 to 17 years old, between 18 to 64 years old, above 65 years old}.

1. *Generate a conversation between a human user and an AI assistant. This human user is a {age} who is {year_range}. Make sure the topic of the conversation or the way that user talks reflects this user's age. You may or may not include the user's age directly in the conversation. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 50% of the dataset)

2. *Generate a conversation between a human user and an AI assistant. This human user is a {age} who is {year_range}. Make sure the topic of the conversation or the way that user talks reflects this user's age. You may or may not include the user's age directly in the conversation. If you include their age, make sure it's a number but not a range. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 50% of the dataset)

A.3 Education

The education is one of {some schooling (elementary school, middle school, or pre-high school), high school education, college and more}.

1. *Generate a conversation between a human user and an AI assistant. The education of this human user is {education}. Make sure the conversation directly or indirectly reflects this user's education level. Be creative on the topics of the conversation. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 66% of the dataset)

2. *Generate a conversation between a human user and an AI assistant. The education of this human user is {education}. Make sure the conversation directly reflects this user's education level. The user may talk about what diploma or academic degree they have during the conversation. Be creative on the topics of the conversation. You can also include daily topic if it can reflect the user's education. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 17% of the dataset)

3. *Generate a conversation between a human user and an AI assistant. The education of this human user is {education}. Make sure the conversation or the user's language directly or indirectly reflects this user's education level. The user may talk about what diploma or academic degree they have during the conversation. Be creative on the topics of the conversation. The topic doesn't have to be academic. You can also include daily topic if it can reflect the user's education. '### Human:', and the AI assistant's response should start with '### Assistant:'* (This instruction was used for generating 17% of the dataset)

A.4 Socioeconomic Status

The socioeco is one of {low, middle, high}. The corresponding class_name is one of {lower, middle, upper}, and the corresponding other_class_name is one of {middle or upper classes, lower or upper classes, lower or middle classes}.

1. *Generate a conversation between a human user and an AI assistant. The socioeconomic status of this human user is {socioeco}. Make sure the conversation reflects this user’s socioeconomic status. You may or may not include this user’s socioeconomic status directly in the conversation. ‘### Human:’, and the AI assistant’s response should start with ‘### Assistant:.’ (This instruction was used for generating 50% of the dataset)*

2. *Generate a conversation between a human user and an AI assistant. The socioeconomic status of this human user is {socioeco}. Make sure the conversation implicitly or explicitly reflects this user belongs to {class_name} class but not {other_class_name}. You may or may not include the user’s socioeconomic status explicitly in the conversation. Be creative on the topic of the conversation. ‘### Human:’, and the AI assistant’s response should start with ‘### Assistant:.’ (This instruction was used for generating 50% of the dataset)*

A.5 System Prompt

When sampling the synthetic conversations from the GPT-3.5-Turbo model, we used the system prompt

“You are a chatbot who will actively talk with a user and answer all the questions asked by the user.”

For the LLaMa2Chat-13B model, we used the following system prompt

“You are a helpful, respectful and honest assistant. Always answer as helpfully as possible, while being safe. Please ensure that your responses are socially unbiased and positive in nature. If a question does not make any sense, or is not factually coherent, explain why instead of answering something not correct. If you don’t know the answer to a question, please don’t share false information.”

B Training details

Trained on Layer 30’s Residual Stream Representations



Figure 5: Effect of training data size on the reading and control probe’s performance. The accuracy is measured on a held-out validation set of each attribute. Probes were trained and validated on the internal representation at 30th layer. In the plots above, the starting training size for gender is 64, for age and socioeconomic status attribute is 80, for education is 90.

We generated 1,000 to 1,500 conversations for each subcategory (e.g. female) of a user attribute (e.g. gender). Our synthetic dataset does **not** contain any duplicated conversations. We used an 80-20